Module B4 — Energy, Reproduction and Populations

4.1 Photosynthesis

Photosynthesis is the process by which light energy is used to convert carbon dioxide and water into glucose and oxygen. It occurs in chloroplasts and is the fundamental process by which energy enters most ecosystems.

$$6CO2 + 6H2O \rightarrow C6H12O6 + 6O2$$

Light-Dependent Reactions

The light-dependent reactions occur on the thylakoid membranes of chloroplasts. They require light energy and produce ATP and reduced NADP (NADPH), which are used in the light-independent reactions.

1. **Photolysis of water:** Light energy splits water molecules: 2H2O → 4H+ + 4e- + O2. The oxygen is released as a waste product.
2. **Photoexcitation:** Light energy is absorbed by chlorophyll in photosystem II (P680), exciting electrons to a higher energy level. These electrons pass along an electron transport chain.
3. **Electron transport chain:** As electrons pass between carrier proteins, they release energy used to pump H+ ions from the stroma into the thylakoid space, creating a proton gradient (chemiosmosis).
4. **ATP synthesis:** H+ ions flow back through ATP synthase (embedded in the thylakoid membrane) down their concentration gradient, driving the synthesis of ATP from ADP + Pi.
5. **Photosystem I (P700):** Electrons from the chain are re-excited by light energy in PSI. These high-energy electrons, combined with H+ ions, reduce NADP to NADPH.

> **Photophosphorylation:** The production of ATP using light energy during the light-dependent reactions.

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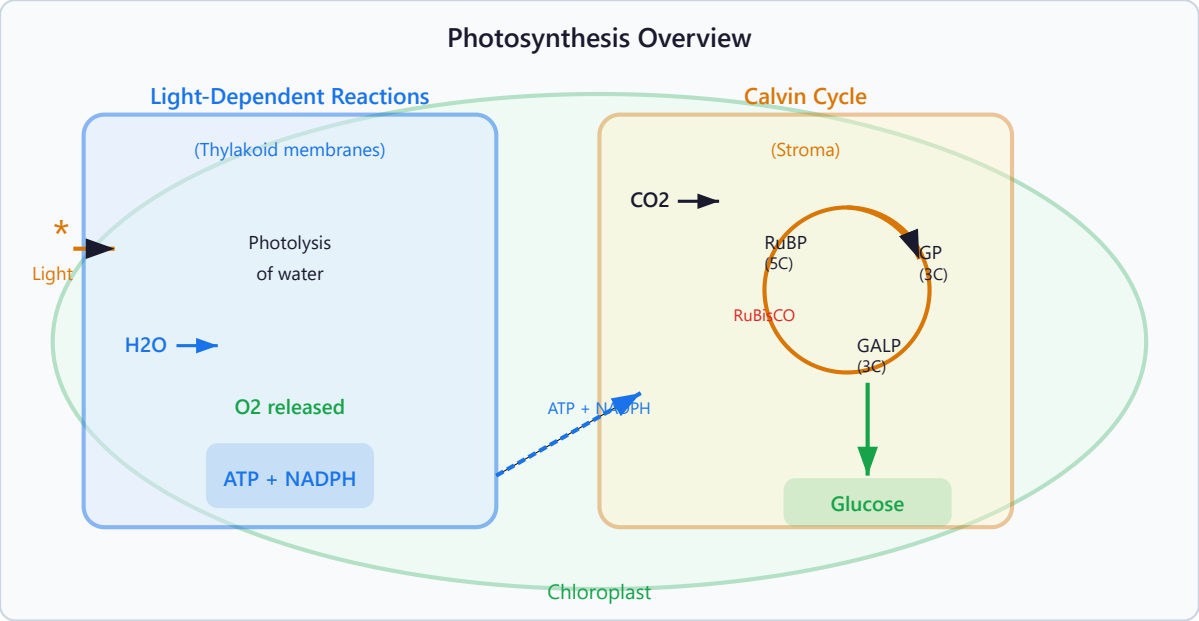
Non-cyclic photophosphorylation: The normal pathway involving both PSI and PSII, producing ATP, NADPH, and O2.

Cyclic photophosphorylation: Electrons from PSI cycle back to the electron transport chain, producing only ATP (no NADPH or O2). Used when more ATP than NADPH is needed.



Light-Independent Reactions (Calvin Cycle)

The Calvin cycle occurs in the stroma of chloroplasts. It uses the ATP and NADPH produced in the light-dependent reactions to fix carbon dioxide into organic molecules.

1. **Carbon fixation:** CO2 combines with ribulose bisphosphate (RuBP, a 5C compound) catalysed by the enzyme RuBisCO, forming an unstable 6C compound that immediately splits into two molecules of glycerate-3-phosphate (GP, 3C).
2. **Reduction:** GP is reduced to glyceraldehyde-3-phosphate (GALP/G3P, 3C) using ATP and NADPH from the light-dependent reactions.
3. **Regeneration of RuBP:** Most GALP molecules (5 out of every 6) are used to regenerate RuBP using ATP. The remaining GALP is used to synthesise glucose and other organic molecules.

For every 3 molecules of CO_2 fixed, one molecule of GALP is produced as net output, requiring 9 ATP and 6 NADPH. To produce one molecule of glucose (6 carbons), the cycle must turn 6 times, fixing 6 CO_2 and using 18 ATP and 12 NADPH.

Limiting Factors of Photosynthesis

The rate of photosynthesis is limited by whichever factor is in shortest supply (the **principle of limiting factors**). At any given time, only one factor limits the rate:

- **Light intensity:** At low light, increasing intensity increases the rate (more photoexcitation, more ATP and NADPH produced). At high light intensity, another factor becomes limiting.
- **CO_2 concentration:** Atmospheric CO_2 (~0.04%) is often limiting. Increasing CO_2 increases the rate of carbon fixation by RuBisCO in the Calvin cycle, up to a plateau when another factor limits.
- **Temperature:** Affects enzyme activity. Below the optimum, increasing temperature increases the rate (more kinetic energy, more enzyme-substrate collisions). Above the optimum, enzymes denature and rate falls sharply. Temperature mainly affects the Calvin cycle enzymes (light reactions are less temperature-sensitive).

Real-world application: Commercial greenhouses optimise all three factors: supplementary lighting extends the photoperiod and increases light intensity; CO_2 enrichment (raising levels to ~0.1%) significantly increases yields; and temperature is controlled at optimal levels (~25-30 degrees C for many crops). This understanding of limiting factors allows growers to maximise photosynthetic rate and crop productivity cost-effectively.

Exam Tip: When interpreting graphs of photosynthesis rate against a limiting factor, the plateau indicates that the factor on the x-axis is no longer limiting — another factor has become limiting. If multiple curves are shown at different levels of a second factor, explain how increasing the second factor raises the plateau, confirming it was the limiting factor at the original level.

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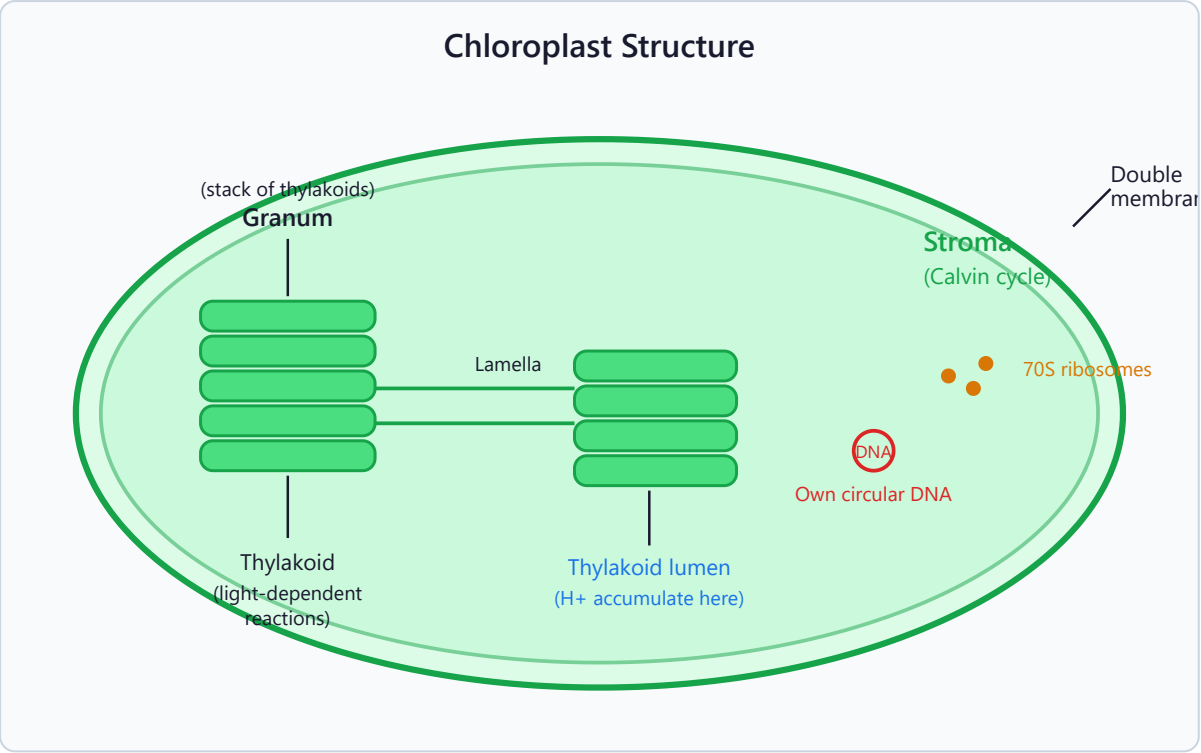

Limiting Factors

The rate of photosynthesis is limited by whichever factor is in shortest supply:

- **Light intensity:** Affects the rate of the light-dependent reactions (photolysis and photoexcitation)
- **CO2 concentration:** Affects the rate of carbon fixation in the Calvin cycle
- **Temperature:** Affects enzyme activity, particularly RuBisCO in the Calvin cycle

Exam Tip: When explaining limiting factors, always state which specific stage or reaction is limited. For example, "Low CO2 limits the rate of carbon fixation by RuBisCO, so less GP is produced, less GALP is formed, and less glucose is synthesised."

4.2 Respiration

Respiration is the metabolic process by which organic molecules (usually glucose) are broken down to release energy, which is used to synthesise ATP. It occurs in all living cells continuously.

Glycolysis

Glycolysis occurs in the cytoplasm and does not require oxygen (anaerobic). It converts one molecule of glucose (6C) into two molecules of pyruvate (3C).

- Glucose is phosphorylated using 2 ATP (activation energy), producing fructose-1,6-bisphosphate (6C)
- This is split into two triose phosphate molecules (3C each)
- Each triose phosphate is oxidised (hydrogen removed and transferred to NAD, producing NADH) and converted to pyruvate
- Net yield per glucose: 2 ATP (substrate-level phosphorylation) + 2 NADH

Link Reaction

In the presence of oxygen, pyruvate enters the mitochondrial matrix. Each pyruvate is decarboxylated (CO_2 removed) and dehydrogenated (hydrogen transferred to NAD), producing acetyl CoA (2C) which enters the Krebs cycle. Per glucose: 2 CO_2 released, 2 NADH produced.

Krebs Cycle

The Krebs cycle occurs in the mitochondrial matrix. Acetyl CoA (2C) combines with oxaloacetate (4C) to form citrate (6C), which is progressively decarboxylated and dehydrogenated in a series of reactions that regenerate oxaloacetate.

Per turn of the cycle (one acetyl CoA): 2 CO_2 released, 3 NADH produced, 1 $FADH_2$ produced, 1 ATP produced (substrate-level). Per glucose molecule (2 turns): 4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP.

Oxidative Phosphorylation

This is the final stage and produces most ATP. It occurs on the inner mitochondrial membrane (cristae).

1. NADH and $FADH_2$ donate electrons to the electron transport chain (series of protein carriers on the inner membrane)
2. As electrons pass along the chain, they release energy used to pump H^+ ions from the matrix into the intermembrane space, creating a proton gradient
3. H^+ ions flow back through ATP synthase (chemiosmosis), driving ATP production
4. At the end of the chain, electrons combine with H^+ and oxygen (the final electron acceptor) to form water

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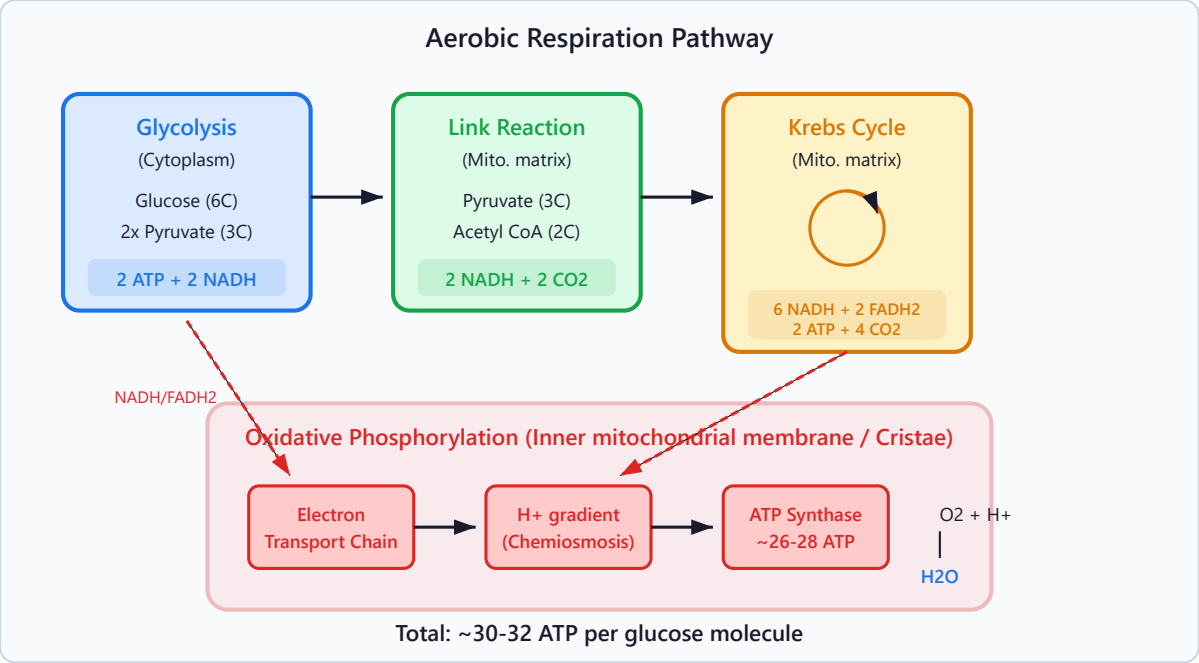
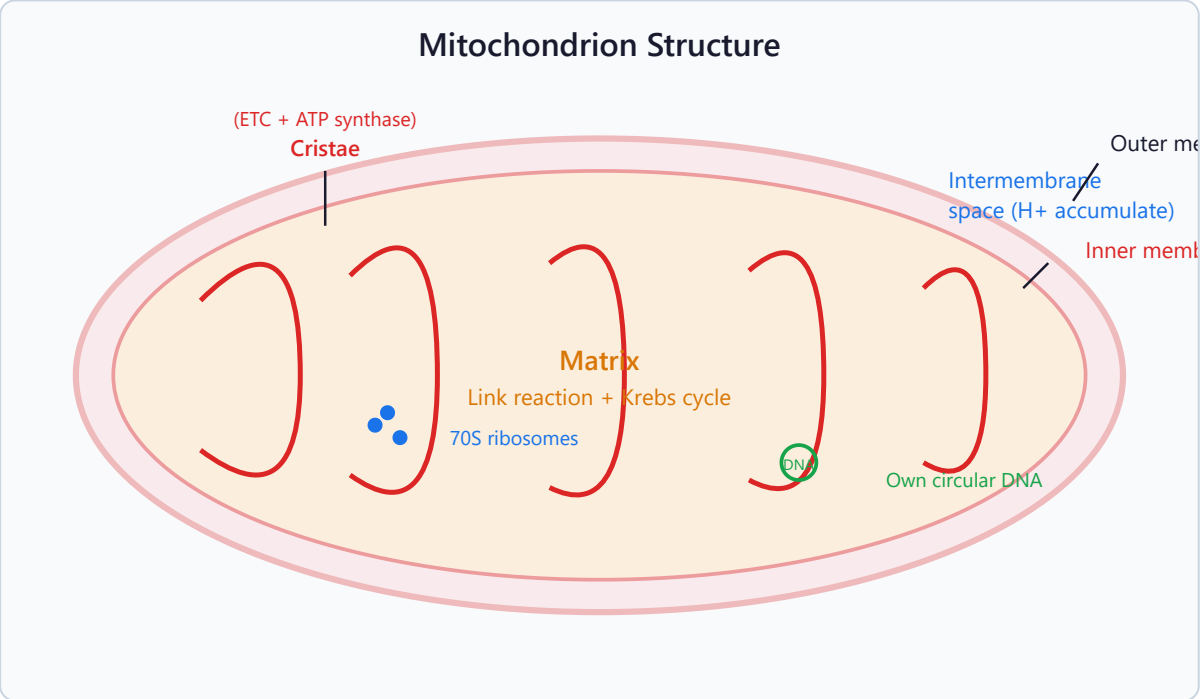
Theoretical maximum ATP yield per glucose: approximately 30-32 ATP (via aerobic respiration)





Anaerobic Respiration

When oxygen is unavailable, oxidative phosphorylation cannot occur and NAD cannot be regenerated. Anaerobic pathways regenerate NAD so glycolysis can continue:

- **In animals:** Pyruvate is reduced to lactate using NADH (lactate fermentation). Reversible when oxygen becomes available.
- **In plants and yeast:** Pyruvate is decarboxylated to ethanal (acetaldehyde), then reduced to ethanol using NADH (alcoholic fermentation). CO2 is also produced. Irreversible.

4.3 Ecosystems and Energy Transfer

Ecosystem: A community of organisms interacting with each other and their abiotic environment in a defined area.
Trophic level: The position of an organism in a food chain, determined by the number of energy transfers from the start of the chain.

Energy Flow

Energy enters most ecosystems as sunlight, which is converted to chemical energy by photosynthesis. Energy flows through trophic levels but is lost at each stage:

- Not all sunlight is absorbed by plants (reflected, wrong wavelength, misses leaves)
- Not all absorbed energy is used in photosynthesis
- Organisms use energy for respiration (released as heat)
- Not all biomass is consumed (roots, bones, etc.)
- Not all consumed material is assimilated (faeces contain undigested material)

$$\text{Net Primary Productivity (NPP)} = \text{Gross Primary Productivity (GPP)} - \text{Respiration (R)}$$

Typically only about 10% of energy is transferred between trophic levels. This limits food chain length (usually 4-5 levels maximum) because insufficient energy remains to support another level.

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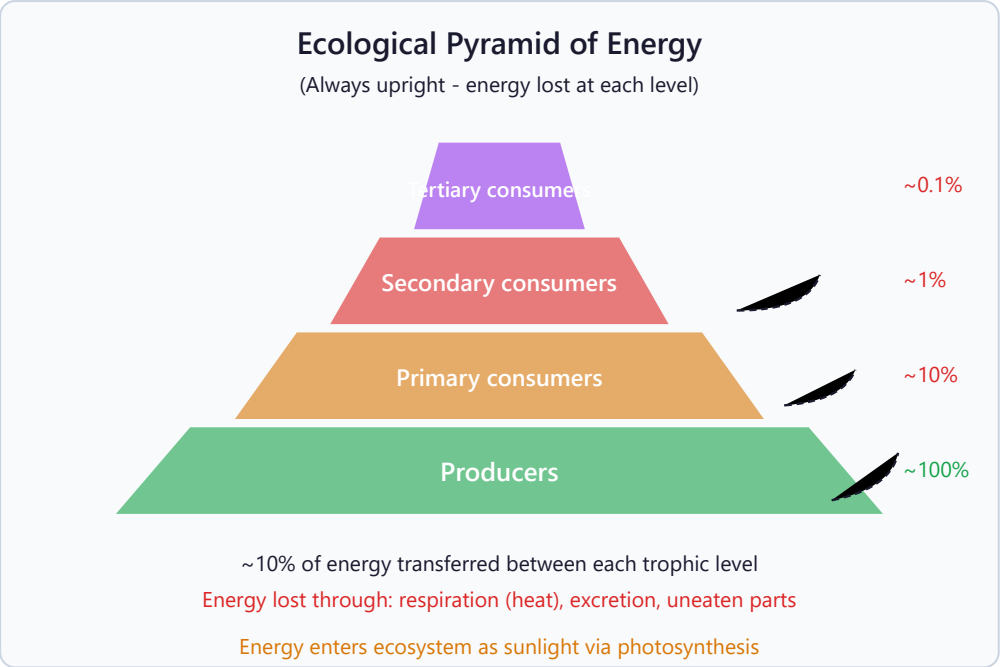


Productivity and Farming

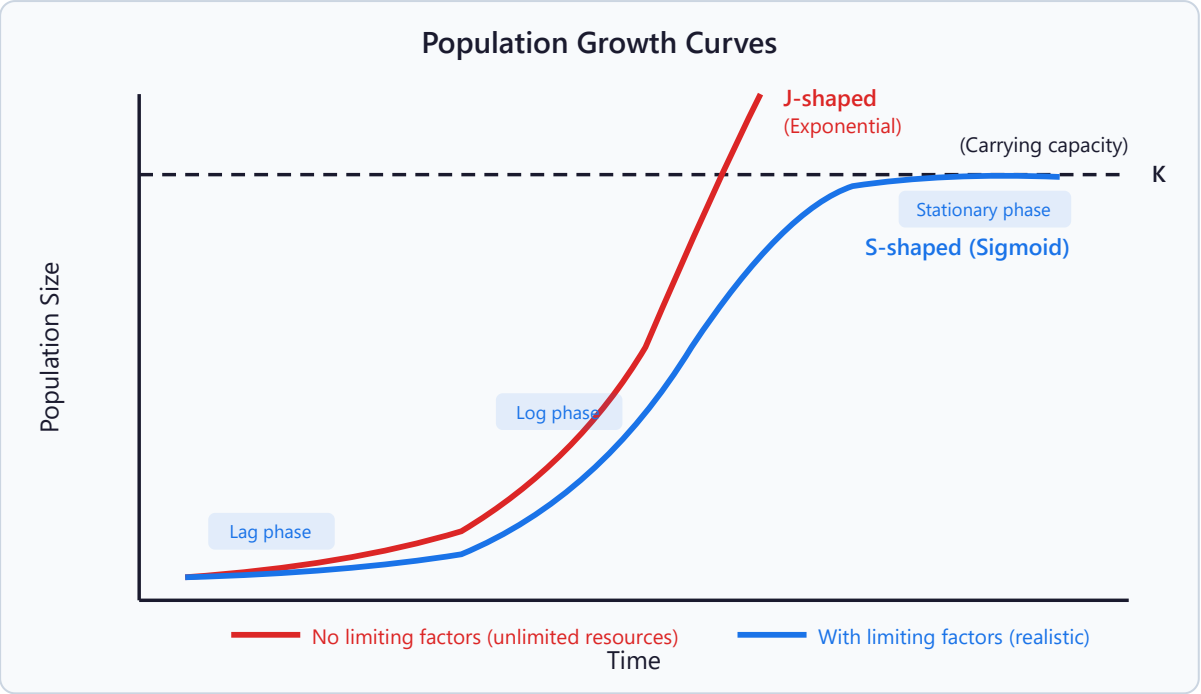
Agricultural practices aim to increase net productivity reaching humans by reducing energy losses. Methods include restricting animal movement (reducing respiratory losses), keeping animals warm (less energy spent on thermoregulation), using pesticides and herbicides (reducing energy lost to competitors and pests), and selective breeding for efficient feed conversion.

4.4 Populations

Population Growth

A population is a group of organisms of the same species living in the same area at the same time. Population growth typically follows a sigmoid (S-shaped) curve:

- **Lag phase:** Slow initial growth as few individuals reproduce and acclimatise to the environment
- **Log (exponential) phase:** Rapid growth as resources are abundant and there is little competition or predation
- **Stationary phase:** Growth slows and population stabilises around the carrying capacity as limiting factors balance birth and death rates



Carrying capacity (K): The maximum population size that an environment can sustain indefinitely given the available resources.
Limiting factors: Environmental factors that restrict population growth, including food, water, shelter, disease, predation, and competition.

Competition

Intraspecific competition (within the same species) is density-dependent and is the main factor regulating population size around carrying capacity. As population density increases, competition for resources intensifies, reducing birth rate and increasing death rate.

Interspecific competition (between different species) occurs when species share the same niche. The competitive exclusion principle states that two species cannot occupy the same niche indefinitely; one will outcompete the other.

Predator-Prey Relationships

Predator and prey populations show linked oscillations: prey numbers increase when predators are few, predator numbers then increase due to abundant food, prey numbers decrease due to increased predation, and predator numbers then decrease due to food shortage. The predator cycle lags behind the prey cycle.

4.5 Reproduction

Sexual and Asexual Reproduction

Sexual reproduction involves the fusion of gametes (fertilisation) from two parents, producing genetically variable offspring. It requires meiosis to produce haploid gametes. Asexual reproduction involves a single parent producing genetically identical offspring (clones) through mitosis, requiring less energy but producing no genetic variation.

4.6 Evolution and Natural Selection

Natural selection: The process by which organisms with heritable characteristics best suited to their environment are more likely to survive, reproduce, and pass on their alleles to the next generation.
Evolution: A change in allele frequency within a population over successive generations.

Natural selection requires: variation within a population (due to mutation and sexual reproduction), environmental selection pressure, differential survival and reproduction (those with advantageous traits produce more offspring), and inheritance of advantageous alleles by the next generation.

Speciation occurs when populations become reproductively isolated (geographically, temporally, or behaviourally), experience different selection pressures, accumulate different genetic changes, and eventually can no longer interbreed to produce fertile offspring.

Exam Tip: When explaining natural selection, always follow the sequence: variation exists, selection pressure acts, individuals with advantageous alleles survive and reproduce more, alleles are passed on, and allele frequency changes over generations. Never say organisms "choose" to adapt or "develop" new features in response to the environment.

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